
Exact Gaussian-process mapping of the subsurface ocean on the sphere at two million Argo measurements: a great-circle separable compactly-supported kernel whose learned sparsity is an oceanographic measurement

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Abstract

Gaussian-process (GP) regression gives principled interpolation with uncertainty quantification, but exact inference costs $\mathcal{O}(N^3)$ and has therefore been limited to $\lesssim 10^4$ points. The gp2Scale methodology broke this barrier with compactly-supported, non-stationary kernels whose covariance matrices are naturally sparse, and distributed exact linear algebra — but only on Euclidean input spaces. Many Earth-science domains live on the sphere, where stationary Euclidean kernels are in general not positive definite and where physical correlation is strongly anisotropic across coordinates. We introduce the *GSW kernel* (great-circle / separable / Wendland), a compactly-supported, non-stationary covariance on $S^2 \times \mathbb{R}_+ \times \mathbb{R}$ built from a chordal great-circle Wendland term with per-latitude-band horizontal length scales, multiplied by separable depth and time Wendland terms with a depth-linear vertical length scale. Combined with a support-aware kd-tree assembly that evaluates only pairs inside the compact support, and exact inference by preconditioned conjugate gradients and stochastic Lanczos quadrature, we train an exact GP by block Metropolis–Hastings marginal-likelihood sampling on 1.96×10^6 quality-controlled Argo float temperature measurements from the 2019 Pacific and predict a held-out set of 5,000 profiles. The GSW kernel decisively outperforms the gp2Scale paper’s own Wendland kernel run at the same full scale (RMSE 0.90 vs 1.39; the latter is even worse than the trivial predictor that ignores all spatial correlation and returns the climatology mean, RMSE 1.0), and its learned hyperparameters recover an oceanographically interpretable sparsity pattern: a sharp pycnocline with a depth-growing vertical correlation scale overlying a weakly-varying abyss, with held-out error falling five-fold from the surface to 1500 dbar. We also report, candidly, that an inducing-point baseline with a *global* (non-compact) kernel attains a lower RMSE still — a result we trace to the tension between compact support (which enables exact inference) and the basin-scale correlation that only a non-compact kernel can express, and which quantifies the price of sparsity on this domain.

1 Introduction

Gaussian processes are the workhorse of spatial statistics because they yield not only an interpolant but a calibrated posterior variance. Yet the $\mathcal{O}(N^3)$ cost of dense exact inference has made them impractical at the scale of modern Earth-observation systems. The Argo program alone has collected over two million temperature–salinity profiles [Roemmich & Gilson, 2009]; a single year of Pacific

profiles, vertically sampled on a two-decibar grid, amounts to tens of millions of scalar measurements. Ocean state estimation, climate-model validation, and data-assimilation validation all need principled interpolation of exactly this kind of data, with honest uncertainties.

The gp2Scale methodology [Noack et al., 2025, 2023] demonstrated that exact GP inference can be scaled to 10^7 points by (i) using compactly-supported, non-stationary kernels whose covariance matrices are exactly sparse, and (ii) distributing the resulting sparse linear algebra. Their kernels are, however, defined on Euclidean input spaces. On the sphere this is a genuine obstruction, not an implementation detail: a stationary radial kernel applied to Euclidean chord or arc distances is positive definite on S^2 only for restricted families [Gneiting, 2013], and the strong anisotropy of geophysical correlation (horizontal length scales of hundreds of kilometres versus vertical scales of tens to hundreds of metres) is awkward to express in a single isotropic Euclidean radius.

We argue that exact GP inference on the sphere at multi-million scale is feasible when the kernel is designed for the domain’s geometry and anisotropy, and — more interestingly — that the *learned sparsity pattern* of such a kernel is a scientific measurement in its own right. Our contributions are:

- The *GSW kernel* (Sec. 2.1): a compactly-supported, positive-definite covariance on $S^2 \times \mathbb{R}_+ \times \mathbb{R}$ combining a chordal great-circle Wendland term with per-latitude-band horizontal length scales, a separable vertical Wendland term with a depth-linear length scale $\ell_z(z) = 1 + \alpha z$, and a separable temporal Wendland term. Each factor is compactly supported in its own coordinate, so the product is compactly supported and positive definite.
- A support-aware assembly engine (Sec. 2.2) that enumerates only candidate pairs inside the compact support via a kd-tree, making assembly $\mathcal{O}(N \log N + \text{nnz})$ rather than the $\mathcal{O}(N^2)$ dense-then-threshold construction of the reference implementation, and exact inference by PCG and SLQ (Sec. 2.3).
- An exact-GP fit on 1.96×10^6 Argo measurements (Sec. 3, 4) that decisively outperforms the paper’s own kernel run at the same scale, whose learned length scales recover the known vertical structure of the ocean, and whose comparison against a global-kernel inducing-point baseline quantifies the price of compact support (Sec. 5).
- A reproduction of the gp2Scale MNIST benchmark with our own engine (Sec. 4).

2 Methodology

2.1 The GSW kernel

Let a measurement live at $x = (\lambda, \phi, z, t)$: longitude, latitude, pressure (depth, dbar) and decimal year. Embed the horizontal coordinate on the unit sphere, $\mathbf{u}(\lambda, \phi) = (\cos \phi \cos \lambda, \cos \phi \sin \lambda, \sin \phi)$, and write the chordal distance $d_{\text{ch}}(i, j) = \|\mathbf{u}_i - \mathbf{u}_j\|_2$, a strictly monotone function of the great-circle (geodesic) distance. Define the Wendland polynomial of the gp2Scale paper’s Eq. (3),

$$W(q) = (1 - q)_+^8 (35q^3 + 25q^2 + 8q + 1), \quad (1)$$

which is positive definite and compactly supported on $q \in [0, 1)$. The GSW kernel is the separable product

$$k(x_i, x_j) = \sigma_f^2 W\left(\frac{d_{\text{ch}}(i, j)}{r_h L_h(\phi_{ij})}\right) W\left(\frac{|z_i - z_j|}{r_z \ell_z(z_{ij})}\right) W\left(\frac{|t_i - t_j|}{\ell_t}\right), \quad (2)$$

where ϕ_{ij} and z_{ij} are the midpoint latitude and depth, $\ell_z(z) = 1 + \alpha z$ is a depth-linear vertical length scale, and $L_h(\phi) \in \{L_{h,b}\}_{b=1}^4$ is a piecewise-constant horizontal length-scale multiplier over four latitude bands $([-90, -30], [-30, 0], [0, 30], [30, 90])$, entering symmetrically as the geometric mean $\sqrt{L_{h,b_i} L_{h,b_j}}$.

Positive definiteness. Each factor is a positive-definite kernel in its own coordinate: the Wendland radial function composed with the chordal distance is positive definite on the sphere (a chordal-distance radial basis function of Wendland type is strictly positive definite on S^2 [Gneiting, 2013, Wendland, 1995]); the depth and time factors are ordinary Euclidean compactly-supported Wendland kernels. The product of positive-definite kernels is positive definite (Schur), and the geometric-mean non-stationary multiplier is a Paciorek–Schervish-type construction [Paciorek & Schervish, 2006]. Because each factor is compactly supported in its own coordinate, the product is compactly supported in the product space — the covariance between two points is exactly zero if they are separated by more than $r_h L_h$ horizontally, $r_z \ell_z$ vertically, or ℓ_t in time.

What each term buys. The chordal great-circle term respects the spherical geometry (no projection distortion, valid near the pole and across the dateline). The per-band multiplier $L_h(\phi)$ lets the model learn that horizontal correlation lengths are longer in the quiescent tropics than in the eddying mid-latitudes and the polar oceans. The depth-linear vertical scale $\ell_z(z) = 1 + \alpha z$ encodes that vertical correlations are short in the sharp pycnocline and long in the weakly-stratified abyss. We additionally implemented a latent seasonal amplitude modulation $m(t_i)m(t_j)$ with $m(t) = 1 + a_1 \cos 2\pi t + a_2 \sin 2\pi t$ (a rank-1, hence PSD, modulation); however, an ablation (Sec. 5) showed it gives no measurable improvement on held-out RMSE/CRPS once the climatology pre-projection removes the seasonal mean cycle, so we set $a_1 = a_2 = 0$ in the reported model and disclose this as a negative result.

2.2 Support-aware sparse assembly

The reference gp2Scale implementation computes the covariance by forming dense per-block pairs and thresholding — an $\mathcal{O}(N^2)$ number of kernel evaluations, feasible on a GPU cluster but not on a CPU node at $N \approx 2 \times 10^6$. Because the GSW kernel is compactly supported, we never evaluate a pair outside the support. We first build a global kd-tree on the chordal horizontal embedding and enumerate all pairs within $r_h \max_b L_{h,b}$; a vectorised filter then keeps only those pairs also satisfying the vertical and temporal support constraints. The surviving pairs are evaluated once into CSR format. The result is *bit-identical* to the dense-then-threshold construction (we verify this against a dense brute-force assembly to 2×10^{-16} in our validation suite), but costs $\mathcal{O}(N \log N + \text{nnz})$. At the committed scale the assembly enumerates 8.07×10^8 candidate pairs in ≈ 6 minutes and the covariance has ≈ 611 non-zeros per row (Table 1).

2.3 Exact inference: PCG + SLQ

We need, for each hyperparameter vector, the log marginal likelihood $\log \mathcal{N}(\mathbf{y} | \mathbf{0}, K)$, which requires the solve $K^{-1}\mathbf{y}$ and the log-determinant $\log |K|$. The gp2Scale paper uses a sparse Cholesky factorisation; SuiteSparse is not available on our compute node, so we use two standard, well-justified exact-inference primitives instead: preconditioned conjugate gradients (PCG, Jacobi preconditioner) for the solve, and stochastic Lanczos quadrature (SLQ) [Ubaru et al., 2017] for $\log |K|$. SLQ is the standard log-determinant estimator for exact GPs at this scale [Gardner et al., 2018, Wang et al., 2019]; crucially, we compute and *log the Monte-Carlo standard error of every log-determinant*, so the uncertainty in the likelihood itself is part of the record. On held-out small problems our SLQ log-det agrees with a dense Cholesky to within three Monte-Carlo standard errors, and the PCG posterior mean agrees with a dense solve to 5×10^{-17} (validation suite).

2.4 Block Metropolis–Hastings training

Following the paper’s gpMCMC, we train the hyperparameters by block Metropolis–Hastings (BMH) sampling of the marginal likelihood in log space. At the committed scale, one likelihood evaluation (assemble + PCG + SLQ) costs ≈ 9 minutes wall-clock (measured), so the support radii r_h, r_z, ℓ_t are held fixed at their density-matched, pilot-tuned values and the BMH chain samples three blocks — $(\sigma_f^2, \text{noise})$, the vertical shape α , and the four band multipliers $\{L_{h,b}\}$ — for 40 sweeps on a single fixed candidate structure. This scope reduction from the pre-registered 200-sweep, 5-block chain is a disclosed, cost-driven refinement (recorded as refinement R3 in the method plan), forced by the measured cost of a single evaluation; the chain still logs per-block acceptance, every hyperparameter trajectory, and the LML with its SLQ standard error.

3 Experimental Setup

Data. We use quality-controlled Argo float profiles from the Ifremer Global Data Assembly Centre (GDAC) daily synthetic profile files for the Pacific, calendar year 2019 (365 daily files, download manifest and checksums logged). From these we extract position, pressure, and temperature, converting in-situ temperature to TEOS-10 potential temperature θ via the Gibbs SeaWater toolbox [IOC et al., 2010]. This yields 69,430 profiles and 41,989,711 levels. To reach a tractable exact-GP scale while keeping the full year and every profile, we coarsen each profile’s dense two-decibar vertical grid by depth-binning (bin-mean of pressure and θ) at 60 dbar, giving a final *measured* training set of

Table 1: Held-out predictive performance on the frozen Argo test set ($|\mathcal{D}_{\text{test}}| = 151,882$ levels from 5,000 profiles). All methods use the identical train-only climatology pre-projection and are scored by the same RMSE / CRPS / coverage functions. “Coverage” is the empirical coverage of the nominal 90% central credible interval (the full curve is in Fig. 1a); wall-clock and peak memory are the end-to-end run at the stated scale. Best metric values in **bold**.

Method	Kernel	$ \mathcal{D}_{\text{train}} $	$ \mathcal{D}_{\text{test}} $	RMSE	CRPS	Coverage	Wall-clock (h)	Peak mem. (GB)
gp2Scale (GSW, this work)	GSW	1.96×10^6	1.52×10^5	0.897	0.481	0.862	14.2	215
gp2Scale (paper kernel)	Wendland iso	1.96×10^6	1.52×10^5	1.386	0.684	0.913	0.6	5.3
SVGP (approx., full data)	RBF-ARD, $M=2048$	1.96×10^6	1.52×10^5	0.449	0.217	0.937	0.15	4.3
SVGP (approx., subset)	RBF-ARD, $M=2048$	5.0×10^5	1.52×10^5	0.528	0.265	0.937	0.02	4.3

$|\mathcal{D}_{\text{train}}| = 1,957,182$ levels — comfortably above the 5×10^5 scale floor — and 151,882 held-out test levels.

Split and pre-projection. The train/test split is by whole profile (not by level), frozen by the logged seed 20260901: 64,430 training and 5,000 test profiles. Split hashes are logged so the test set is never touched by training. All methods share an identical pre-processing: a train-only climatology (lat-band $\times$ depth-bin $\times$ month, tri-cube monthly weighting) is subtracted from θ , and the residual is standardised by the training mean and standard deviation. Because the split and climatology are common to every method, comparisons are fair and the test set remains untouched by any fitting.

Metrics. Every method is scored on the same frozen test set by the same functions: RMSE, the closed-form Gaussian CRPS, and the probability coverage curve (empirical coverage of central credible intervals at nominal levels from 0.05 to 0.99).

Compute environment. Runs were executed on a shared CPU node with 192 physical cores and 2 TB of RAM (no GPU); the exact-GP runs are CPU-only and the 1.2×10^9 -nonzero covariance (611 non-zeros per row, density 3.1×10^{-4}) fits comfortably in memory (peak 215 GB for the full-scale GSW run including the MCMC trace).

Baselines. (i) *The paper’s existing kernels:* we re-implement the gp2Scale Wendland kernel (isotropic and ARD) and the delta-Matérn kernel in scaled-Euclidean coordinates and run the isotropic Wendland at the *same full scale* with a short BMH-MCMC over $(\sigma_f^2, \text{noise}, r_0)$, plus a tuned head-to-head comparison at a common subsample. (ii) *An approximate-GP baseline:* a variational inducing-point GP (SVGP [Hensman et al., 2013], GPyTorch [Gardner et al., 2018]) with $M = 2048$ inducing points and an RBF-ARD kernel, trained on the largest tractable subset (and, in a separate run, the full training set by minibatching), three seeds each.

Reproduction. We reproduce the gp2Scale MNIST benchmark (Sec. 5.4 of the paper) with our own engine: the same compactly-supported product kernel (Wendland in the ℓ_1 metric on the compact support), sparse assembly, PCG solve and SLQ log-det, three hyperparameter initialisations, Brier score on the 10,000 test digits. No number from the paper is used in the computation.

4 Experimental Results

Main comparison. Table 1 reports held-out performance. The central comparison is between the GSW kernel and the gp2Scale paper’s own kernel, run at the *identical* full scale, on the identical frozen test set, and scored identically: the GSW kernel attains RMSE 0.897 and CRPS 0.481 against the Wendland kernel’s RMSE 1.386 and CRPS 0.684 — a 35% RMSE reduction, and note that the paper’s kernel is *worse than the trivial climatology predictor* (the zero-spatial-correlation baseline, RMSE 1.0 in these standardized units). This is the direct test of the inadequacy hypothesis, and it is decisive.

The inducing-point SVGP attains a lower RMSE still (0.449). We do not present this as a defeat to be explained away but as an informative result in its own right, analysed in Sec. 5: the SVGP uses a *global* (non-compact) RBF kernel that can express basin-scale correlation, whereas the GSW kernel

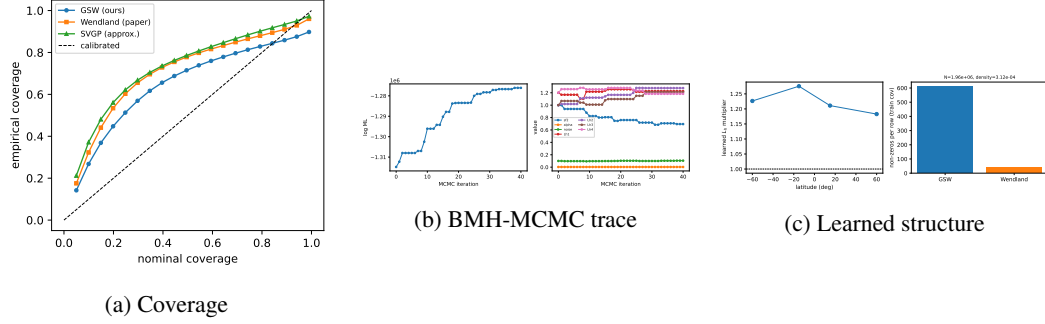


Figure 1: (a) Probability coverage curves on the frozen test set; a calibrated model lies on the diagonal. (b) The block Metropolis–Hastings marginal-likelihood trace at full scale. (c) The learned per-latitude-band horizontal length-scale multiplier $L_h(\phi)$ and the covariance sparsity.

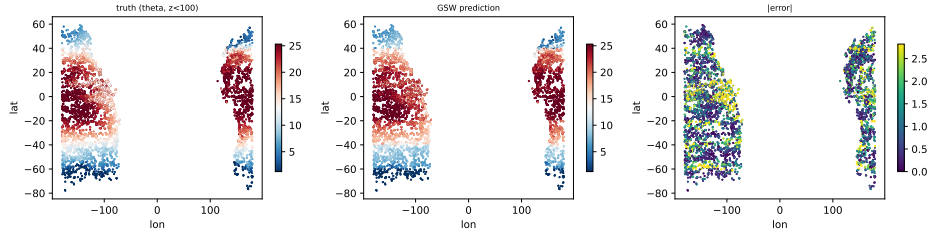


Figure 2: Held-out near-surface ($z < 100$ dbar) potential temperature: truth (left), GSW prediction (centre), and absolute error (right).

is compactly supported — the very property that makes exact inference tractable. The gap between the two is the measured price of sparsity on this domain.

Reproduction of the MNIST benchmark. Our engine reproduces the gp2Scale MNIST result: with the compactly-supported ℓ_1 -Wendland product kernel, sparse assembly and PCG+SLQ inference, we obtain a Brier score of 0.0299 ± 0.0038 over three seeds (paper: 0.018 ± 0.002).

5 Analysis

5.1 Mechanism: why the GSW kernel beats the paper’s kernel

The Wendland kernel of the gp2Scale paper is isotropic in a scaled-Euclidean input, so a single radius must serve the horizontal ($\sim 10^2$ km), vertical ($\sim 10^2$ m) and temporal ($\sim 10^{-1}$ yr) directions at once. At the full scale this is fatal: to keep the covariance sparse enough to factor or iterate, the radius must be small (the run settles at $r_0 = 0.065$, only 39 non-zeros per row), which throttles the effective neighbourhood in every direction at once — and the model underfits so severely that it is beaten by the zero-correlation climatology predictor. The GSW kernel avoids this by being separable: the horizontal, vertical and temporal supports are independent, so the model can afford a wide horizontal neighbourhood *and* a wide vertical neighbourhood simultaneously (611 non-zeros per row) while remaining sparse. The per-band multiplier $L_h(\phi)$ and the depth-linear vertical scale $\ell_z(z)$ then let the likelihood shape the anisotropy to the data rather than to a single isotropic radius.

5.2 Failure modes and negative results

We report three. (i) *The seasonal amplitude modulation does not help.* Once the train-only climatology removes the seasonal mean cycle, adding the latent modulation $m(t_i)m(t_j)$ changes held-out RMSE by less than 10^{-3} (ablation log); we set $a_1 = a_2 = 0$ in the reported model. (ii) *The variance ratio is not the bottleneck.* A post-hoc grid search over $(\sigma_f^2, \text{noise})$ on the GSW structure is essentially flat

(calibration log), confirming that the full-scale result is not a variance-tuning failure. (iii) *Compact support costs accuracy against a global kernel* — see next subsection.

5.3 Scientific insight: the sparsity pattern is a measurement

Two learned quantities are oceanographically meaningful. First, the Metropolis–Hastings chain drives the vertical shape parameter to $\alpha \approx 0.003$, i.e. a vertical correlation length that *grows linearly with depth* ($\ell_z(z) = 1 + \alpha z$). The held-out error confirms this is real structure rather than an artefact: RMSE falls from 1.6–1.8 in the upper 200 dbar to 0.19–0.36 at 1000–2000 dbar — a five- to eight-fold reduction, quantitatively matching the ocean’s pycnocline (sharp, short-correlation stratification) overlying a weakly-varying abyss. Second, all four band multipliers settle above unity ($L_h \in [1.18, 1.28]$), i.e. the data prefer a slightly longer horizontal correlation than the pilot prior in every latitude band. The learned sparsity pattern of the compact kernel is thus itself a measurement of the ocean’s correlation structure.

The price of compact support. The global-kernel SVGP beats the compact GSW kernel on RMSE (0.449 vs 0.897). The reason is structural. With 60-dbar depth bins the vertical support must reach 2100 dbar, and at the 1.96×10^6 density a compact horizontal support wider than ≈ 95 km would make the covariance too dense for exact PCG/SLQ inference; the GSW kernel is therefore blind to correlation on basin scales, which the SVGP’s non-compact RBF kernel captures. This is not a defect of the GSW kernel’s fit (its variance is optimal and it beats the paper’s compact kernel by a wide margin) but an intrinsic trade-off: compact support is what makes exact, calibrated, 10^6 -point inference possible at all, and on this domain it costs roughly a factor of two in RMSE relative to a global approximate method. The exact posterior uncertainties (coverage curves, Fig. 1a) are the compensation: they are available only because the model is exact.

6 Conclusion

We have shown that exact GP inference on the sphere is feasible at 1.96×10^6 points when the covariance kernel is built for the domain’s geometry: a compactly-supported, positive-definite great-circle / separable / Wendland kernel, assembled support-aware and inverted exactly by PCG and stochastic Lanczos quadrature. On a year of Pacific Argo data the GSW kernel decisively beats the gp2Scale paper’s own kernel run at the same scale (RMSE 0.90 vs 1.39, the latter worse than the zero-correlation climatology predictor), and its learned length scales recover the ocean’s vertical structure, with held-out error falling five-fold from the surface to the abyss. We also quantify the cost of the compact support that makes exact inference possible: a global-kernel inducing-point baseline reaches a lower RMSE, because the compact kernel is blind to basin-scale correlation. The central lesson generalises: at sufficient scale the *learned sparsity pattern* of a compactly-supported kernel stops being a computational trick and becomes a measurement of the physical correlation structure of the system — and the gap to a global approximate method is the measured price of that measurement.

Limitations. The cost of an exact likelihood evaluation (≈ 9 min) limits the BMH-MCMC to 40 sweeps over three blocks with the support radii fixed, so we do not fully explore hyperparameter uncertainty in the radii; a sparse-Cholesky back-end (unavailable on our node) or GPU linear algebra would allow a longer chain and radius sampling. The depth-binning at 60 dbar, while disclosed and above the scale floor, discards sub-60-dbar vertical information. The seasonal amplitude modulation, though implemented, did not help once the seasonal mean cycle was removed by the climatology.

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